PHC6050 Statistical Methods for Health Science I (3 credit hours)

Spring 2026

Delivery Format: Online (asynchronous)

Open Access Course Materials: Biostat Open Learning Textbook (BOLT)

Course website: Canvas

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Office Hours: To be posted
See Canvas page for Zoom link.

Preferred Course Communications:

- Ask about specific questions or issues of a personal nature by email through the Canvas inbox in E-learning.
- Ask more general questions (NOT personal or specific quiz questions) on the discussion board in E-learning.

Prerequisites

There are no specific prerequisite courses, but students should be comfortable working with equations, performing basic mathematical calculations (e.g., order of operations, fractions, and square roots), and working with computers.

Purpose and Outcome

Course Overview

Statistical methods for description and analysis provide investigators with useful tools for making sense of data. The pervasiveness of statistics in public health as well as other fields has led to increased recognition that statistical literacy—familiarity with the goals and methods of statistics—should be a basic component of a well-rounded educational program. In this course, students will develop statistical vocabulary, learn methods for descriptive data analysis, study the fundamentals of probability and sampling distributions, learn methods for statistical inference and hypothesis testing based on one or two samples, and become familiar with categorical data analysis and linear regression. Data analysis will be conducted in SPSS.

Relation to Program Outcomes

This three-credit course is a required concentration core course for MPH Biostatistics students and covers the following competencies.

- Describe the role of biostatistics in public health research.
- Use appropriate statistical methodology to address public health problems.
- Apply software to conduct statistical analyses.
- Interpret and critique analyses found in public health studies.

Course Objectives and/or Goals

Upon successful completion of this course, students will be able to

- CO-1: Describe the role biostatistics serves in the discipline of public health.
- CO-2: Differentiate among different sampling methods and discuss their strengths and limitations.

- CO-3: Describe the strengths and limitations of designed experiments and observational studies.
- CO-4: Distinguish among different measurement scales, choose the appropriate descriptive and inferential statistical methods based on these distinctions, and interpret the results.
- CO-5: Determine preferred methodological alternatives to commonly used statistical methods when assumptions are not met.
- CO-6: Apply basic concepts of probability, random variation, and commonly used statistical probability distributions.
- CO-7: Use statistical software to analyze public health data.
- CO-8: Develop presentations based on statistical analyses for both public health professionals and educated lay audiences.

Instructional Methods

This is a fully online asynchronous course. All lecture slides, pre-recorded lectures/demo videos, quizzes, homework assignments, course projects, discussion board, announcements, and course messaging system, are available in E-learning. An independent/auxiliary open learning resource (including textbook and software tutorial videos) is also free for you to access either from weekly module on our Canvas course website or directly at BOLT.

Description of Course Content

Topical Outline

Week	Date(s)	Topic(s)
1	1/12	Introduction and Preliminaries Unit 1A – EDA for One Variable
2	1/19	Unit 1B – EDA for Two Variables
3	1/26	Unit 2 – Producing Data
4	2/2	Unit 3A – Probability
5	2/9	Unit 3B1 – Discrete Random Variables
6	2/16	Unit 3B2 – Continuous Random Variables
7	2/23	Unit 3B3 – Sampling Distributions
8	3/2	Unit 4A1 – Estimation
9	3/9	Unit 4A2 – Hypothesis Testing
SB	3/16	Spring Break
10	3/23	Unit 4B1 – Inference for Relationships (Case CQ)
11	3/30	Unit 4B1 – Inference for Relationships (Case CQ)
12	4/6	Unit 4B2 – Inference for Relationships (Case CC)
13	4/13	Unit 4B3 – Inference for Relationships (Case QQ)
14	4/20	Wrap-Up
15	4/27	Final Exam Week

A detailed schedule of assignments with due dates can be found at the end of the syllabus.

Course Materials and Technology
All course materials are available online either through **CANVAS** or in the **online open access textbook** (Biostat Open Learning Textbook (BOLT)).

There is no required textbook to purchase for this course. However, the following textbooks may serve as useful references with additional examples/exercises:
- Daniel, W.D. (2013): Biostatistics: A Foundation for Analysis in the Health Sciences. 10th Edition, Wiley.
- Agresti, A. (2013): The Art and Science of Learning from Data. 4th Edition, Pearson.

IMPORTANT: You will also need access to the statistical software package, SPSS. Course materials discuss multiple software packages. **In PHC 6050 you are only responsible for SPSS**.

Videos, Recorded Lectures & Demos
Most videos presented in the course material are stored in YouTube. If the text in the video is too blurry, try increasing the quality of the YouTube video using the small gear icon which appears at the bottom of the video when it is playing. If you want to view the video faster or slower, you can adjust the speed using the gear icon. Many videos have closed captions and/or transcripts available. In addition to these videos, recorded lectures and demos will be posted at the start of each week. Lectures will largely recap course readings while demos will allow me to walk through examples for you.

Statistical Software – SPSS
Students can access SPSS by purchasing SPSS license or via UFApps for free.

- There are two ways of SPSS Licensing for Students at UF (You can also obtain this software from other resources).
    - Direct student leasing is available for use on personally owned computer. It can be purchased ON CAMPUS at the UF Computing Help Desk located at the 132 HUB Stadium Road (or search for "SPSS Licensing for Students" at https://software.ufl.edu/). The most current version is SPSS v31. SPSS is available for both Windows and Macintosh. It may require additional time for the OS X media to be created. Please let the Help Desk know if you require additional media for a non-Windows install. Currently, it is not possible to obtain this software remotely (off main campus).
    - All UF students located off main campus are directed to use IBM® SPSS® educational sales program with ONTHEHUB. It is currently the only means to provide this media.

  Note: The version available through UF is less expensive than the versions you can buy elsewhere; however, if you buy this software from another source be sure to obtain either the STANDARD GRADUATE PACK (GRADPACK) or PREMIUM GRADUATE PACK. The BASE or other versions may not have enough functionality for this course.
- SPSS is also available on UFApps for students. Besides SPSS, this free Apps server also provides various other applications such as Microsoft Office. This works for any computing device from any location at any time, if you log in with your GatorLink Credentials. Many students have been successful at using this system for their assignments in this course. I highly recommend storing your files on the M: drive (can also be found in UFApps) because it provides the best performance when working with files in UFApps.

SPSS Tutorials
Tutorials for all skills needed for assignments in this course can be found on BOLT. We have tried to make it as easy as possible to follow along with the tutorials. Watching the videos at a slower speed can help. Viewing the transcripts while you watch or work in SPSS may also help.

Whenever possible, many students find it helpful to have the videos playing in one window, monitor, or other device while working in the software in another, pausing as needed to work through the process with your own data.

Recommended SPSS Books (optional – extra resources)

Although we will provide tutorials for all SPSS skills required for this course, there are numerous SPSS guides available if you wish to purchase one. The best for you may depend on what you might be doing with SPSS after our course. I have had recommendations from students for Julie Pallant's "SPSS Survival Manuals." Many resources are available both in print and online via the UF library. Your recommendations for others are also appreciated.

E-learning

An E-Learning site is available for the course (http://elearning.ufl.edu). **The weekly schedule and all course materials are available online through this site including grades, assignments, discussions boards, and other course information. It is very important to check the weekly page and review all announcements carefully.**

Academic Requirements and Grading

Quizzes

There will be untimed quizzes in Canvas covering the material assigned for the review of the previous week. Quizzes test basic definitions and skills and may sometimes be cumulative in that they will go back and ask earlier questions. Quizzes are typically due on Friday at 11:59 pm (some quizzes may due on Tuesday at 11:59pm). You have up to three attempts for each quiz before its due date and your highest of these scores will be recorded. Quizzes test basic definitions and skills and may sometimes be cumulative in that they will go back and ask earlier questions. We highly recommend that you start your first attempt early and take your three attempts on different days with time for reviewing the course material in between. When you submit a quiz attempt, you will see your grade and will be able to review your quiz attempt. For each question, you will see whether you answered correctly or incorrectly but it will not reveal the correct answer for any you did not answer correctly. There will also be feedback for each question which will direct you to the most important content to review.

Assignments

Most assignments will involve data analysis in software and interpretation and/or certain questions which cannot be easily presented in the quizzes. Assignments will normally be due on Tuesdays at 11:59 pm (some may be due on Fridays), but all assignments except the first (Pre-semester Survey) and last (Post-semester Survey) will require extended work and should be started as early as possible, no later than the week prior to the due date, in order to have time to address any questions or issues. For all software assignments in this course, if you do not receive full credit for the software part (Part A), you may resubmit Part A before the deadline for Part B to receive half credit back for your corrections.

Course Project

Each student will individually perform a guided data analysis based upon two (hopefully linearly related) quantitative variables. These variables will then be categorized in two ways (2 levels, 3+ levels). The relationship between the two variables will be investigated using different combinations of variable types. This course project will be completed in 4 steps during the semester.

All assignments must be submitted via E-Learning by the exact due date and time. All work you submitted together with our comments can be reviewed by clicking Grades on the navigation panel in Canvas.

Grading

Requirement	% of final grade
Quizzes (14)	15%
Assignments (8)	40%
Software for Assignments (4)	15%
Software for Course Project (Steps 1, 2, 3)	15%
Course Project Step 4	15%

Points Earned	[93,100]	[90,93)	[87,90)	[83,87)	[80,83)	[77,80)	[73,77)	[70,73)	[67,70)	[63,67)	[60,63)	<60
Letter Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	E
Grade Points	4.0	3.67	3.33	3.0	2.67	2.33	2.0	1.67	1.33	1.0	0.67	0

Please be aware that a C- is not an acceptable grade for graduate students. The GPA for graduate students must be 3.0 based on 5000 level courses and above to graduate. A grade of C counts toward a graduate degree only if based on credits in courses numbered 5000 or higher that have been earned with a B+ or higher. More information on UF grading policy may be found at: http://gradcatalog.ufl.edu/content.php?catoid=10&navoid=2020#grades

Policy Related to Make up Work

All work must be submitted via E-Learning by the exact due date and time. Any late submission or missed work will receive a grade of ZERO unless arrangements have been made ahead of the due date with the instructor. Late submissions or make-ups are acceptable ONLY due to illness or other unanticipated circumstances warranting a medical excuse and resulting in the student missing an assignment deadline, consistent with College policy. Documentation from a health care provider is required. Please note: Any requests for make-ups due to technical issues MUST be accompanied by UF Computing help desk (http://helpdesk.ufl.edu/) correspondence. You MUST e-mail me within 24 hours if you wish to request a make-up.

Policy Related to Required Class Attendance

This is an online course. "Attendance" means you are expected to go through the course materials, take notes, and pay attention to and post in the discussion boards. This is to be done at your own pace, but assignments and quizzes have scheduled deadlines to keep you on track. You are welcome to get ahead if you need flexibility in future weeks. Requirements for class attendance and make-up exams, assignments, and other work in this course are consistent with university policies that can be found at https://catalog.ufl.edu/ugrad/current/regulations/info/attendance.aspx

ACADEMIC POLICIES & RESOURCES

Information about University Academic Policies and Resources.

Student Expectations, Roles, and Opportunities for Input

Expectations Regarding Course Behavior

It's critical to review the weekly page in Canvas and read all announcements carefully. Each week's materials will be clearly identified on the course E-learning site. Students are expected to work through the material as scheduled. It is very important to work through all content contained on this site as directed and ask questions about the material you do not understand. **Working through the content from start to finish is the best approach to achieve a high level of understanding and success in this course.** In addition, it is your responsibility to review the comments and feedback we give on your graded assignments.

Communication Guidelines

Questions about course material should be asked during office hours or posted on the course discussion boards in E- Learning. Questions about specific quiz questions or issues of a personal nature should be sent by email through E-Learning. For questions asked Monday-Thursday, we will try our best to respond within 24 hours. For questions asked Friday-Sunday, we will respond Monday or as soon as possible thereafter.

ADVICE

All that I can ask is that you do your best with the provided materials and to ask when you need more direction or explanation. It is expected that you will spend approximately 10-12 hours per week on this course. Scheduling your time wisely and working efficiently will minimize the need for extra work in this course. Generally, I advise students to break this time up into blocks of 1-3 hours split over as many days of the week as possible given your schedule. Working on too much material in one sitting is more likely to cause frustration and does not allow for time for understanding to develop or for questions to be answered. Learn to use the materials to your greatest advantage. There is a lot of content, but if you understand the examples or if you have experience with certain topics, it may not be necessary to review all of the content we provide. The questions presented in the "Learn by Doing" and "Did I Get This" activities as well as the course worksheets are indicative of important questions and concepts that you will need to understand and are designed to teach as well as test your understanding. We highly encourage you to go through these as they are presented in the online textbook (for the "Learn by Doing" and "Did I Get This" activities) and the course E-learning page (for the worksheets). If you go through the content as directed, you will learn the skills you need to succeed in the course as well as build a foundation of statistical knowledge. If you ever feel lost, please ask, but also understand that the course is building to a complete picture, and it's sometimes hard to see how each topic is related until later in the semester when we tie everything together. Often, activities and worksheets are leading you to think about things that will be important later in the course while working on skills related to the current topic. Watch the software tutorials carefully, especially if you find the software aspect challenging, and review our suggestions in the SPSS information section. Do not allow yourself to waste time working in the software. If you are having issues, let us know immediately and we will help as soon as possible. Try to make sure as much of your time as possible in the software is productive. Be sure to ask when you don't understand, and stay on track with the material. Getting behind can be difficult to fix in any course, but especially an asynchronous one which is largely self-guided. Let the instructor know as soon as possible if you feel you are falling behind.

PHHP STUDENT RESOURCES

PHHP's UPTurn (Unified Pathways to Support Wellness) program is a *no-cost mental health and wellness program* that is offered year-round to all PHHP students (undergraduate, graduate and professional level) and students (from any college) who are enrolled in PHHP courses. UPTurn advisors support students on their wellness journeys by curating individualized plans (resources and support) to help them manage academic, social, emotional, and health-related stress.

Interested students are paired with an UPTurn advisor, who meets with each student *virtually* (Zoom, Teams, phone) or *in person* (private office/room in HPNP) for a 45-minute consultation, followed by (if desired):

1. Up to 4 follow-up skills-building visits
2. When needed and appropriate, up to 10 psychotherapy sessions after completion of the 4 follow-up skills-building visits

Note: UPTurn is NOT a crisis/emergency resource. Students who are in crisis are strongly encouraged to use UF's existing crisis support resources.

Students can learn more about UPTurn as well as request an appointment.

Any questions regarding UPTurn can be directed to upturn@phhp.ufl.edu or (352) 273-6850.

Inclusive Learning Environment

Public health and health professions are based on the belief in human dignity and on respect for the individual. As we share our personal beliefs inside or outside of the classroom, it is always with the understanding that we value and respect diversity of background, experience, and opinion, where every individual feels valued. We believe in, and promote, openness and tolerance of differences in ethnicity and culture, and we respect differing personal, spiritual, religious and political values. We further believe that celebrating such diversity enriches the quality of the educational experiences we provide our students and enhances our own personal and professional relationships. We embrace The University of Florida's Non-Discrimination Policy, which reads, "The University shall actively promote equal opportunity policies and practices conforming to laws against discrimination. The University is committed to non-discrimination with respect to race, creed, color, religion, age, disability, sex, sexual orientation, gender identity and expression, marital status, national origin, political opinions or affiliations, genetic information and veteran status as protected under the Vietnam Era Veterans' Readjustment Assistance Act." If you have questions or concerns about your rights and responsibilities for inclusive learning environment, please see your instructor or refer to the Office of Multicultural & Diversity Affairs website: www.multicultural.ufl.edu